

Attribute Controllable Beautiful Caucasian Face Generation by Aesthetics Driven Reinforcement Learning

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EigenGAN



Our Approach

Figure 1: Top: the original faces generated by EigenGAN [3]. Bottom: aesthetically improved ones by aesthetics-driven reinforcement learning over semantic attributes of the generator.

ABSTRACT

In recent years, image generation has made great strides in improving the quality of images, producing high-fidelity ones. Also, quite recently, there are architecture designs, which enable GAN to unsupervisedly learn the semantic attributes represented in different layers. However, there is still a lack of research on generating face images more consistent with human aesthetics. Based on EigenGAN [He et al., ICCV 2021], we build the techniques of reinforcement learning into the generator of EigenGAN. The agent tries to figure out how to alter the semantic attributes of the generated human faces towards more preferable ones. To accomplish this, we trained an aesthetics scoring model that can conduct facial beauty prediction. We also can utilize this scoring model to analyze the correlation between face attributes and aesthetics scores. Empirically, using off-the-shelf techniques from reinforcement learning would

not work well. So instead, we present a new variant incorporating the ingredients emerging in the reinforcement learning communities in recent years. Compared to the original generated images, the adjusted ones show clear distinctions concerning various attributes. Experimental results using the MindSpore, show the effectiveness of the proposed method. Altered facial images are commonly more attractive, with significantly improved aesthetic levels.

ACM Reference Format:

Xin Jin, Shu Zhao, Le Zhang, Xin Zhao, Qiang Deng, and Chaoen Xiao. 2022. Attribute Controllable Beautiful Caucasian Face Generation by Aesthetics Driven Reinforcement Learning. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3503161.3547737>

1 INTRODUCTION

In recent years, Generative Adversarial Networks (GANs) [2] show promising performance on image synthesis. A parallel line of research tries to learn interpretable GAN generators, which can discover explicit dimensions controlling the semantic attributes. InfoGAN in [1] learns disentangled representations by decomposing the conventional single unstructured noise vector into incompressible noise and a set of structured latent variables. Motivated by style transfer literature, Karras et al. proposed StyleGAN in [4], enabling separations of high-level attributes like pose and identity.

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MM '22, October 10–14, 2022, Lisboa, Portugal
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ACM ISBN 978-1-4503-9203-7/22/10.
<https://doi.org/10.1145/3503161.3547737>

Most recently, He et al. proposed EigenGAN [3], which for each generator layer directly embeds in one linear subspace model with an orthogonal basis. During the training, EigenGAN automatically learns a set of “eigen-dimensions” corresponding to interpretable semantic attributes. So until now, people have several ways to manipulate the semantics within the GANs, or said in other words, semantically edit them. One problem followed is how to enhance the generated face images positively.

In this paper, we incorporate the techniques of reinforcement learning (RL) into the attribute values searching in EigenGAN. The first requirement of the RL agent is the design of the reward function. The objective is to enhance the images to the ones more aesthetically attractive. So for this, we trained an aesthetics score network based on SCUT-FBP5500 released in [5]. By this, we give a multi-attribute model using MindSpore[8], which can conduct automatic Caucasian face beautification for the generated images, as shown in Fig. 1. The user can select different facial attributes in the interaction section to change the generated image.

2 EIGENGAN + RL

2.1 Preliminaries for RL

In RL, the agent constantly interacts with the environment, and this process is described as a Markov Decision Process (MDP). The objective is to learn a stochastic policy, such that when taking actions, the expected reward of the trajectories is maximized. In this paper, the discounting factor equals 1. Also, no rewards exist for intermediate states. The agent only gets a reward for the terminal state. The reward value is the aesthetic score of the image after a series of adjustments.

2.2 Generator of EigenGAN

The controlling parameters of the generator consist of not only the conventionally noise input $\epsilon \in \mathcal{N}(0, \mathbf{I})$, but also a sequence of latent variables $\mathbf{z}_i \in \mathbb{R}^q$, which are similarly sampled from $\mathcal{N}_q(0, \mathbf{I})$ (for EigenGAN, q is set to 6). Six blocks of transposed convolution layers constitute the main part of the neural architecture. For the input part, it is a fully connected layer mapping ϵ into a $4 \times 4 \times 512$ image. Correspondingly for the output, it is passed into a convolutional layer, getting a generated 256×256 image of 3 channels.

2.3 Learn to set semantic attributes

For EigenGAN, dimensions from latent variables $\mathbf{z}_i$ correspond to interpretable or controllable semantic attributes. So as we move along these dimensions, the generator produces a sequence of images with continuous changes corresponding to the relevant semantic attributes. Six layers in EigenGAN brings us six such $\mathbf{z}$'s. We only adopt a subset of these controllable semantic attributes, as the others are a bit entangled or drastic. We first set the dimensions corresponding to the attributes as zero. The agent learns how to select this dimension during the training process. After these setting zero operations, we concatenate these altered $\mathbf{z}_i$'s into one single vector. Let it be $\mathbf{y}$. We clone the single vector five times. The value of five caters to the five trajectories or decision sequences sampled by the RL agent. Intuitively, we train the agent over these sampled trajectories. The agent then gradually learns to select the preferred one.

2.4 Soft policy gradient + self-critical

The standard objective in reinforcement learning is the expected sum of rewards. Most RL algorithms are working on the conventional notion of deterministic optimality. However, intuitively the conventional objective does not consider the importance of stochasticity for exploration. In this paper, we instead follow the framework of maximum entropy reinforcement learning (see e.g. [12]). Also, we follow the approach proposed in [9], i.e., incorporating the self-critical way of training into the soft policy gradient updates. More concretely, the empirical estimate $\hat{Q}_{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t)$'s are replaced with Monte Carlo estimations over single trajectories. These estimations correspond to the final five aesthetics scores given by the aesthetics score network. This disposes the need for estimations via separate neural networks for predicting $\hat{Q}_{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t)$'s. State-dependent baseline is set as the average of the five scores. This baseline deviates slightly from [9] (which uses the baseline score got by the greedy policy under current policy π_θ) but experimentally it is more effective as shown in [7]. In a word, the main difference from conventional self-critical training is applying the way of self-critical on the soft (not the vanilla) policy gradient.

2.5 Aesthetics score network

In this paper, we designed a network dedicated to scoring the aesthetics of human faces. We use the lightweight module Efficient Channel Attention (ECA) [11], and we adopt EfficientNet-B4 [10] as a pre-trained model to extract features of the image. Then a convolution operation is performed with a kernel size of 3. We normalize the features before using the ReLU activation function. Next, it is input to the ECA module, and we flatten it into a 1-dimensional vector after activation and adaptive average pooling. After this, it is input into the regression network, which returns the aesthetics score. We trained this aesthetic face scoring model above using the SCUT-FBP5500 dataset [5].

2.6 Face attribute selection

We use Spearman's rank correlation coefficient or Spearman's ρ measures the dependence between facial semantic attributes and aesthetic quality as in [6]. We have to choose the dimensions with better decoupling effects, i.e., one only observes continuous changes of a *single* semantic attribute when traversing the coefficients of those dimensions. To further narrow the scope, only the attributes with strong correlation ($|\rho| > 0.3$) and also being significant ($p\text{-value} < 0.01$) are selected. After this step, we choose only five semantics, including body pose, bangs, lighting, gaze, and lipstick color.

ACKNOWLEDGMENTS

This work is partially supported by the National Natural Science Foundation of China (62072014 & 62106118), the CAAI-Huawei MindSpore Open Fund (CAAI-XSJLJ-2021-022A), the Open Fund Project of the State Key Laboratory of Complex System Management and Control (2022111), the Project of Philosophy and Social Science Research, Ministry of Education of China (20YJC760115), and the Advanced Discipline Construction Project of Beijing Universities (20210051Z0401). We gratefully acknowledge the support of MindSpore, CANN and Ascend AI Processor used for this research.

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